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Research Paper On,

“Outlier Detection in Cultivation Costs”

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CERTIFICATE

This is to certify that Ms. Divya Lengule of class MSc-II Statistics, Has satisfactorily completed project on “**Outlier Detection in Cultivation Costs**” in the subject-Project (Code: STS-681-RP) during the academic year 2025-2026 and has same been to examined by the guide.

Project Guide

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Abstract

This study presents a comprehensive statistical and time series analysis of agricultural cultivation costs and crop yield patterns to understand variability, underlying trends, and economic relationships in agricultural production systems. The analysis begins with descriptive statistical measures, which reveal substantial heterogeneity and instability in cultivation costs across different crops, years, and input components. Measures of skewness and kurtosis indicate that the distribution of agricultural cost data is highly non-normal, characterized by heavy tails and pronounced extreme values. These findings are further supported by systematic outlier detection using statistical techniques, highlighting significant anomalies in input cost structures.

To examine temporal dynamics, time series analysis is conducted on key agricultural variables. The results show an overall upward trend in cultivation costs, accompanied by intermittent fluctuations in crop yields, suggesting the influence of external factors such as climatic variability, policy changes, and market conditions. Stationarity testing using the Augmented Dickey-Fuller (ADF) test and KPSS test confirms that most variables are non-stationary in nature, necessitating appropriate transformations such as differencing to ensure model reliability.

Furthermore, a regression analysis is employed to investigate the relationship between cultivation cost and crop yield. The results indicate a statistically meaningful association, suggesting that increased investment in agricultural inputs can positively influence productivity, although the relationship may not be strictly linear due to inefficiencies and external constraints.

Overall, the study highlights pronounced variability, structural inefficiencies, and economic stress within the agricultural sector. The findings emphasize the importance of optimized resource allocation, risk management strategies, and data-driven decision-making to improve agricultural productivity and sustainability in the long term.

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INTRODUCTION

Agriculture plays a vital role in the economy by ensuring food security, generating employment, and supporting rural livelihoods. However, the sector is characterized by significant variability in production costs due to differences in crops, regions, and time periods. Understanding the distribution and behavior of cultivation costs is therefore essential for identifying economic stress and improving resource allocation.

Cost of cultivation data often exhibits irregular patterns, including extreme values that deviate substantially from typical observations. These extreme values, known as outliers, are not merely statistical anomalies but may reflect real-world conditions such as adverse weather events, inefficient input usage, regional disparities, or market fluctuations. Identifying such outliers is crucial for detecting underlying economic stress and understanding structural inefficiencies in agriculture.

This study focuses on the detection and analysis of outliers in cost of cultivation data across different crops, regions, and time periods. Various statistical techniques, including Interquartile Range (IQR), Z-score, and robust methods such as Median Absolute Deviation (MAD), are applied to systematically identify unusual observations. The analysis further incorporates descriptive statistics and distributional measures to understand variability and skewness in the data.

In addition, time series techniques are used to examine trends and patterns over time, while stationarity tests such as the Augmented Dickey-Fuller Test and KPSS Test help assess the stability of the data. To explore the relationship between cultivation cost and productivity, regression analysis is performed, and forecasting is carried out using the ARIMA Model.

The main objective of this study is to identify patterns of economic stress and variability in agriculture and provide insights that can support better decision-making for farmers, policymakers, and stakeholders. By distinguishing between normal variation and significant anomalies, the study contributes to a deeper understanding of cost dynamics in the agricultural sector.

Importance of Agricultural Cost Analysis:

Agricultural cost analysis is essential for understanding the economic viability and sustainability of farming systems. It helps in:

- Evaluating the profitability of different crops
- Identifying cost inefficiencies and resource misallocation
- Supporting policy decisions and subsidy planning
- Understanding the impact of external shocks such as inflation, droughts, and market fluctuations
- Improving farm-level decision-making and productivity

Overall, cost analysis combined with outlier detection provides valuable insights into financial risks, economic stress, and sustainability challenges in agriculture, enabling better planning and informed decision-making.

OBJECTIVES

- To identify anomalies in agricultural cost variables using IQR, Z-score, and MAD methods for robust outlier detection.
- To perform component-wise analysis of input costs such as labour, seeds, fertilizers, and irrigation to detect variations and irregularities.
- To compare outliers across different states in order to understand regional disparities in agricultural cost structures.
- To analyze year-wise variations in outliers to identify temporal trends and the impact of external shocks on agricultural costs.
- To examine the relationship between cost outliers and crop yield to understand the cost–yield linkage.
- To identify economically stressed crops and classify the nature of stress as structural, environmental, or policy-driven.

LITERATURE REVIEW

Previous Studies on Agricultural Cost Analysis:

Several studies in agricultural economics have focused on analyzing the cost of cultivation and its impact on productivity and farm efficiency. Research shows that agricultural costs are influenced by multiple factors such as labor, fertilizers, irrigation, and machinery inputs. Many studies highlight that rising input costs often lead to reduced profit margins and increased financial pressure on farmers.

Recent systematic reviews in agriculture emphasize the importance of **data-driven approaches** for understanding cost variability and improving decision-making in farm management. Studies also show that cost structures vary significantly across regions, crops, and time periods, indicating high heterogeneity in agricultural systems. Additionally, modern research increasingly uses statistical and machine learning techniques to analyze cost patterns and forecast future trends in agriculture.

Outlier Detection in Economic Data:

Outlier detection plays an important role in economic and agricultural data analysis, as extreme values can significantly affect statistical results and interpretations. Outliers in agricultural cost data often arise due to unusual weather conditions, sudden changes in input prices, policy shocks, or measurement errors.

Research in statistical data analysis shows that methods such as IQR, Z-score, and robust techniques like MAD are widely used for identifying abnormal observations. Studies also indicate that outliers are not always errors; in many cases, they represent **real economic shocks or extreme agricultural conditions**.

Recent literature further highlights that removing or properly analyzing outliers improves model accuracy in regression and forecasting models. In time series and economic studies, outlier-aware models provide better predictions and more reliable policy insights.

RESEARCH GAP

The reviewed studies collectively highlight the importance, trends, and challenges associated with the cost of cultivation in Indian agriculture. The Manual on Cost of Cultivation Surveys provides a standardized framework for collecting and analyzing cost data, defining cost concepts (A1 to C2) and ensuring consistency across regions and time. Building on this foundation, empirical studies show that the cost of cultivation has increased significantly over time, mainly driven by rising input prices, particularly labour, which contributes the largest share of total cost. While there has been a gradual shift from traditional inputs to mechanization, substitution between labour and machinery remains limited, and technological progress in terms of yield improvement has been relatively slow. Additionally, increasing costs of fertilizers, irrigation, and seeds, combined with inelastic demand for inputs, have further escalated production expenses.

These rising input costs have adversely affected farm profitability, as output prices have not increased at the same pace, leading to a cost–price imbalance. Although government interventions such as subsidies and Minimum Support Prices (MSP) provide some support, they are often insufficient, especially for small and marginal farmers who face constraints in accessing modern technology and resources. Overall, the studies emphasize that rising input costs, limited technological advancement, and structural inefficiencies are key factors affecting agricultural sustainability. They suggest that a balanced approach involving cost control, improved access to inputs, mechanization, and stronger policy support is essential for enhancing farm profitability and ensuring long-term agricultural development.

DATA DESCRIPTION

Source of Data:

The dataset used in this study is based on agricultural cost of cultivation data collected over multiple years for different crops and regions in India. The data is obtained from secondary sources such as government agricultural surveys and research databases related to farming economics.

The dataset includes detailed information on input costs, production expenses, crop yield, and temporal variations across different years. It is used to analyze long-term trends and detect abnormal cost behavior in agriculture.

DATASET URL:

<https://ckandev.indiadataportal.com/dataset/cost-of-cultivation/resource/be22e37e-acfc-43a9-859c-d4a066489a69>

Description of Variables (Cost Components, Yield, etc.):

The dataset consists of both cost-related and production-related variables. The major variables used in this study include:

- Cost of Cultivation Components:
 - Human labour cost
 - Machine labour cost
 - Seed cost
 - Fertilizer cost
 - Irrigation cost
 - Total cultivation cost (e.g., C2 cost)
- Output Variable:
 - Crop yield (production per unit area)
- Temporal Variable:
 - Year of cultivation

These variables help in understanding the relationship between input costs and agricultural productivity. The total cost variable is mainly used for outlier detection and economic stress analysis.

Data Structure (Crop, Region, Year-wise):

The dataset is structured in a panel format, where observations are recorded across multiple dimensions:

- **Crop-wise structure:** Different crops such as cereals, pulses, and cash crops are included.
- **Region-wise structure:** Data is collected from multiple geographical regions/states, capturing regional variation in agricultural practices.
- **Year-wise structure:** The dataset spans multiple years, allowing analysis of temporal trends and cost fluctuations over time.

This multi-dimensional structure enables comparative analysis of cost behavior across crops, regions, and time periods.

Data Cleaning and Preprocessing:

Before analysis, the dataset was carefully cleaned and preprocessed to ensure accuracy and consistency. The following steps were performed:

- **Handling Missing Values:** Missing observations in cost and yield variables were treated using appropriate imputation or removal techniques depending on their frequency.
- **Outlier Checking:** Extreme values were initially retained for analysis, as they are important for identifying economic stress conditions.
- **Data Type Conversion:** All numerical variables were converted into appropriate formats for statistical analysis.
- **Standardization of Variables:** Units of measurement were checked to ensure consistency across all records.
- **Filtering and Structuring:** Irrelevant or incomplete records were removed, and the dataset was organized into crop-wise, region-wise, and year-wise formats.

This preprocessing ensured that the dataset was suitable for statistical modeling, outlier detection, and time series analysis.

DATA ANALYSIS

Descriptive Statistical Analysis:

Descriptive statistics such as mean, median, standard deviation, minimum, maximum, and coefficient of variation (CV) are computed for all cost of cultivation variables.

mean	median	std	cv_percent	skewness	kurtosis	
id	1844.000000	1844.00	1065.066899	57.758509	0.000000	-1.200000
state_code	19.335321	22.00	9.493293	49.098192	-0.332422	-1.241837
crop_code	447.715912	203.00	401.747490	89.732681	0.639904	-1.414786
cul_cost_a1	20782.602120	15014.81	20482.419263	98.555605	2.755171	11.162229
cul_cost_a2	21060.124175	15223.31	20666.973319	98.133198	2.707654	10.767752

Interpretation:

The results show **high variability in cultivation costs across crops and years**. The large standard deviation and high CV indicate that agricultural production costs are not uniform. This reflects differences in input usage, farming practices, and regional conditions. The presence of wide ranges between minimum and maximum values suggests strong heterogeneity in agricultural cost structures.

Skewness and Kurtosis Analysis:

Skewness measures symmetry of distribution, while kurtosis measures the presence of extreme values. These are computed using statistical libraries.

skewness	kurtosis	
id	0.000000	-1.200000
state_code	-0.332422	-1.241837
crop_code	0.639904	-1.414786
cul_cost_a1	2.755171	11.162229
cul_cost_a2	2.707654	10.767752
...	...	...
fix_cost_land_rev_tax_cess	6.367701	57.318861
fix_cost_depr_impl_farm_build	2.706413	14.125179
fix_cost_interest_fix_cap	4.006505	29.232048
fix_cost	2.504562	9.152901

Interpretation:

Most variables exhibit **positive skewness**, indicating that a small number of observations have very high cultivation costs. Kurtosis values are significantly high, indicating **heavy-tailed distributions**. This confirms the presence of extreme cost values, suggesting that agricultural data is not normally distributed and is influenced by extreme economic conditions.

Outlier Detection Techniques:

IQR Method:

The Interquartile Range (IQR) test is a statistical method used to identify outliers in a dataset by focusing on the spread of the middle 50% of the data. It is based on quartiles, where the first quartile (Q1) represents the 25th percentile and the third quartile (Q3) represents the 75th percentile. The IQR is calculated as the difference between Q3 and Q1, which captures the central dispersion of the data while ignoring extreme values. Any observation that lies significantly below or above this range is considered a potential outlier.

Outliers are identified using:

$$IQR = Q3 - Q1$$

Data points outside $[Q1 - 1.5 \times IQR, Q3 + 1.5 \times IQR]$ are considered outliers.

Why IQR test use?

The IQR test is widely used because it is a robust method that is not affected by extreme values or skewed distributions. Unlike methods that rely on the mean and standard deviation, such as the Z-score, the IQR method uses median-based measures, making it more reliable for real-world data that often contain irregularities. In the context of agricultural cost analysis, where data may vary due to regional differences, climate conditions, or unexpected economic shocks, the IQR method helps in accurately detecting unusually high or low cost values without being distorted by extreme observations. This makes it especially useful for identifying anomalies in cost components like labor, fertilizers, or irrigation, allowing researchers to better understand inconsistencies and underlying patterns in the dataset.

Output:

	year	crop_name	cul_cost_c2
15	2023-24	Paddy	113266.74
111	2023-24	Groundnut	116117.76
115	2023-24	Groundnut	134221.17
117	2023-24	Groundnut	142380.27
159	2023-24	Cotton	106421.19
...	...	...	...
1885	2013-14	Sugarcane	133840.60

Interpretation:

The IQR method detects several extreme cost values across crops. These outliers represent unusual increases in cultivation costs due to factors such as weather shocks, input price spikes, or inefficient resource usage.

Z-score Method

Z-score measures how far a data point is from the mean in standard deviation units. Values beyond ± 3 are considered outliers.

In practical terms, the Z-score helps identify whether a value is typical or unusual compared to the rest of the data. A Z-score close to zero indicates that the value is near the mean, while a large positive or negative Z-score suggests that the value is far from the average. Typically, values with Z-scores greater than +3 or less than -3 are considered outliers, although this threshold may vary depending on the context of the analysis.

Why Z score method use?

The Z-score test is widely used because it provides a simple and effective way to detect outliers and understand the distribution of data. In agricultural cost analysis, for example, it helps in identifying unusually high or low cost values that may arise due to factors such as extreme weather conditions, market fluctuations, or data entry errors. By detecting such anomalies, researchers can ensure more accurate analysis and better interpretation of economic patterns. Additionally, the Z-score is useful for comparing observations across different regions, crops, or time periods, since it removes the effect of scale and variability in the data.

Output:

	<u>year</u>	<u>crop_name</u>	<u>cul_cost_c2</u>
117	2023-24	Groundnut	142380.27
164	2023-24	Onion	162732.58
166	2023-24	Onion	143481.01
167	2023-24	Onion	209323.61
168	2023-24	Onion	203077.63
...	...	...	...
1576	2015-16	Sugarcane	170931.00

Interpretation:

The Z-score method confirms the presence of extreme values identified in the IQR method. These extreme points indicate abnormal cost behavior in certain crops and years, reflecting economic stress conditions.

MAD Method

Median Absolute Deviation (MAD) is used as a robust measure of variability and is less sensitive to extreme values.

The Median Absolute Deviation (MAD) is a robust statistical measure used to detect variability in a dataset and identify outliers. Unlike methods that rely on the mean and standard deviation, MAD is based on the median, which makes it much less sensitive to extreme values. To compute MAD, you first calculate the median of the dataset, then find the absolute deviations of each observation from this median, and finally take the median of those deviations. This results in a measure of spread that reflects the typical distance of data points from the center without being distorted by unusually large or small values.

Why MAD method use?

The MAD test is widely used in outlier detection because real-world data, especially in fields like agricultural cost analysis, often contain skewed distributions and extreme observations. Methods such as the Z-score can be heavily influenced by outliers since they depend on the mean and standard deviation. In contrast, MAD provides a more reliable and stable approach in such situations because it resists the influence of extreme values. By comparing each observation's deviation from the median relative to the MAD, it becomes easier to flag unusually high or low values as outliers.

Output:

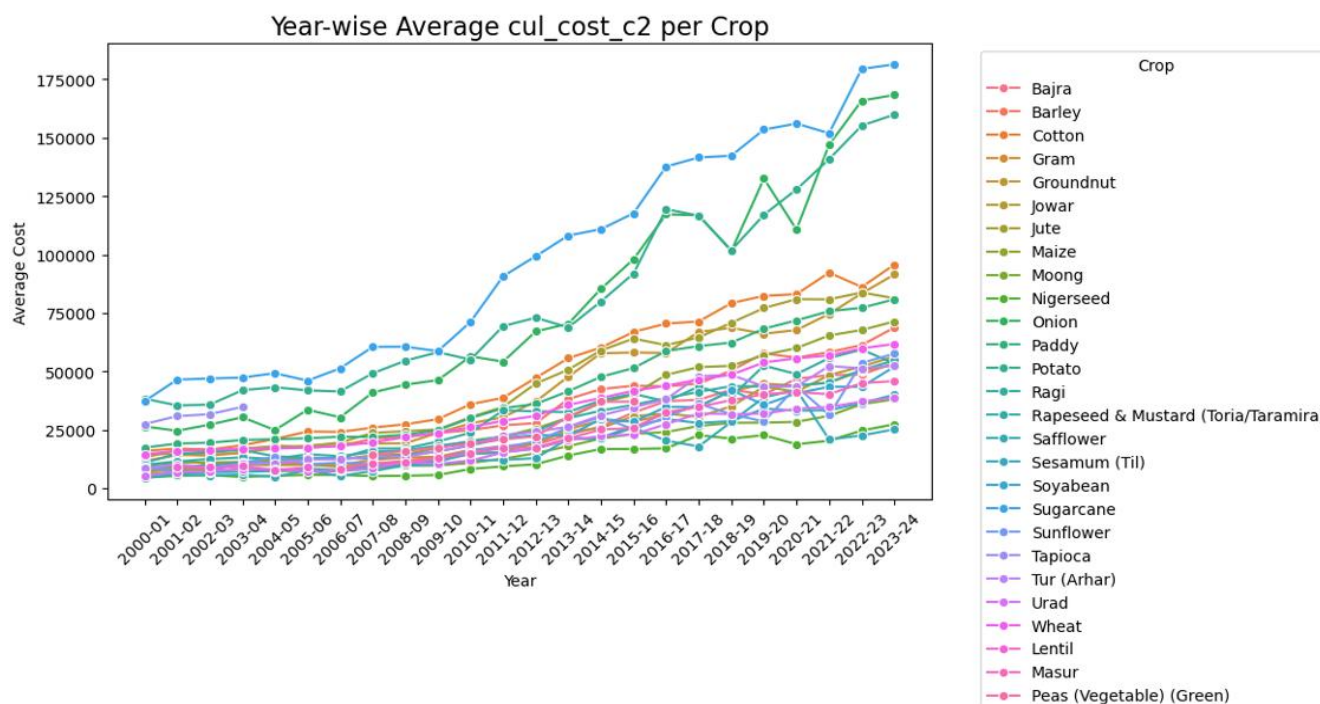
	<u>year</u>	<u>crop_name</u>	<u>cul_cost_c2</u>	<u>mad_flag</u>
<u>111</u>	<u>2023-24</u>	<u>Groundnut</u>	116117.76	1
<u>115</u>	<u>2023-24</u>	<u>Groundnut</u>	134221.17	1
<u>117</u>	<u>2023-24</u>	<u>Groundnut</u>	142380.27	1
<u>161</u>	<u>2023-24</u>	Cotton	124492.93	1
<u>163</u>	<u>2023-24</u>	Onion	136576.01	1
<u>164</u>	<u>2023-24</u>	Onion	162732.58	1
<u>166</u>	<u>2023-24</u>	Onion	143481.01	1
<u>167</u>	<u>2023-24</u>	Onion	209323.61	1
<u>168</u>	<u>2023-24</u>	Onion	203077.63	1

Interpretation:

The MAD method identifies outliers more effectively in skewed data. It confirms that extreme cost observations are not random errors but **systematic deviations caused by real agricultural and environmental factors**.

Time Series Analysis:

Time series plots and trend analysis are used to study cost behavior over time.



Interpretation:

The analysis shows an **increasing trend in cultivation costs over the years**, along with fluctuations. This indicates rising agricultural input costs and unstable economic conditions. Yield patterns also show variability, suggesting that productivity is influenced by external factors such as climate and input availability.

Stationarity Testing:

1. Augmented Dickey-Fuller (ADF) Test:

The Augmented Dickey-Fuller Test (ADF test) is a widely used statistical method in time series analysis to determine whether a given series is stationary or contains a unit root. In simple terms, stationarity means that the statistical properties of a series—such as mean, variance, and autocorrelation—remain constant over time. Many economic and agricultural variables, like cultivation costs or crop yields, often show trends or fluctuations over time, making it essential to test for stationarity before applying further models.

The ADF test is an extension of the basic Dickey-Fuller test. It “augments” the original test by including lagged differences of the variable in the model to account for higher-order autocorrelation. This makes the test more reliable when dealing with real-world data, where observations are often correlated with their past values. The test works by estimating a regression equation and checking whether the coefficient of the lagged level of the series is significantly different from zero.

Null hypothesis (H_0) :states that the time series has a unit root, meaning the series is **non-stationary**.

Alternative hypothesis (H_1): states that the time series does not have a unit root, meaning the series is **stationary**.

	<u>crop</u>	<u>ADF Statistic</u>	<u>p-value \</u>		<u>Critical Value (1%)</u>	<u>Critical Value (5%)</u>	<u>Critical Value (10%) \</u>	<u>Conclusion</u>
0	Paddy	<u>2.566845</u>	<u>0.999068</u>	0	-4.012034	-3.104184	-2.690987	0 <u>Non-Stationary</u>
1	Maize	<u>2.467999</u>	<u>0.999039</u>	1	-3.752928	-2.998500	-2.638967	1 <u>Non-Stationary</u>
2	Jowar	<u>2.057444</u>	<u>0.998743</u>	2	-4.012034	-3.104184	-2.690987	2 <u>Non-Stationary</u>
3	Bajra	<u>0.881814</u>	<u>0.992848</u>	3	-4.012034	-3.104184	-2.690987	3 <u>Non-Stationary</u>
4	Ragi	<u>1.709080</u>	<u>0.998151</u>	4	-4.012034	-3.104184	-2.690987	4 <u>Non-Stationary</u>
5	Wheat	<u>1.069686</u>	<u>0.994949</u>	5	-3.964443	-3.084908	-2.681814	5 <u>Non-Stationary</u>
6	Barley	<u>1.865451</u>	<u>0.998470</u>	6	-3.809209	-3.021645	-2.650713	6 <u>Non-Stationary</u>
7	Tur (Arhar)	<u>1.654929</u>	<u>0.998013</u>	7	-4.012034	-3.104184	-2.690987	7 <u>Non-Stationary</u>
8	Moong	<u>1.131573</u>	<u>0.995477</u>	8	-3.788386	-3.013098	-2.646397	8 <u>Non-Stationary</u>
9	Urad	<u>0.467238</u>	<u>0.983843</u>	9	-3.964443	-3.084908	-2.681814	9 <u>Non-Stationary</u>
10	Gram	<u>-0.384627</u>	<u>0.912634</u>	10	-4.012034	-3.104184	-2.690987	10 <u>Non-Stationary</u>
11	Masur	<u>1.820165</u>	<u>0.998388</u>	11	-4.331573	-3.232950	-2.748700	11 <u>Non-Stationary</u>
12	Groundnut	<u>1.813306</u>	<u>0.998375</u>	12	-3.964443	-3.084908	-2.681814	12 <u>Non-Stationary</u>
13	Soyabean	<u>1.918382</u>	<u>0.998557</u>	13	-3.769733	-3.005426	-2.642501	13 <u>Non-Stationary</u>
14	Sunflower	<u>1.978275</u>	<u>0.998644</u>	14	-4.012034	-3.104184	-2.690987	14 <u>Non-Stationary</u>
15	Sesamum (Til)	<u>1.481696</u>	<u>0.997455</u>	15	-3.788386	-3.013098	-2.646397	15 <u>Non-Stationary</u>
16	Rapeseed & Mustard	<u>2.425220</u>	<u>0.999022</u>	16	-3.964443	-3.084908	-2.681814	16 <u>Non-Stationary</u>
17	Nigerseed	<u>0.481339</u>	<u>0.984295</u>	17	-3.752928	-2.998500	-2.638967	17 <u>Non-Stationary</u>
18	Safflower	<u>-2.822329</u>	<u>0.055154</u>	18	-4.012034	-3.104184	-2.690987	18 <u>Non-Stationary</u>
19	Cotton	<u>-1.382672</u>	<u>0.590554</u>	19	-4.012034	-3.104184	-2.690987	19 <u>Non-Stationary</u>
20	Onion	<u>-1.530568</u>	<u>0.518339</u>	20	-4.012034	-3.104184	-2.690987	20 <u>Non-Stationary</u>
21	Potato	<u>-4.861109</u>	<u>0.000041</u>	21	-4.012034	-3.104184	-2.690987	21 <u>Stationary</u>
22	Sugarcane	<u>-1.069870</u>	<u>0.726937</u>	22	-3.889266	-3.054358	-2.666984	22 <u>Non-Stationary</u>
23	Jute	<u>0.148850</u>	<u>0.969242</u>	23	-3.752928	-2.998500	-2.638967	23 <u>Non-Stationary</u>
24	Peas (Vegetable) (Green)	<u>2.228862</u>	<u>0.998905</u>	24	-5.354256	-3.646238	-2.901198	24 <u>Non-Stationary</u>

Interpretation:

If the p-value ≤ 0.05 , we reject the null hypothesis, meaning the data is stationary.

If the p-value > 0.05 , we fail to reject the null hypothesis, meaning the data is non-stationary and may require differencing or transformation.

Since the majority of crops fail the stationarity condition, the overall conclusion is that the dataset is **non-stationary in nature**, meaning agricultural cost series are influenced by trends, external shocks, or time-varying factors

2. KPSS Test:

The **Kwiatkowski–Phillips–Schmidt–Shin (KPSS) Test** is a statistical test used in time series analysis to check whether a series is **stationary**. In simple terms, stationarity means that the statistical properties of a series—such as its mean, variance, and autocorrelation—remain constant over time. This concept is very important when working with economic or agricultural data, because many analytical models, especially time series models, assume that the data is stationary before making reliable inferences or forecasts.

What makes the KPSS test different from other stationarity tests, like the Augmented Dickey-Fuller Test, is its hypothesis structure. In the KPSS test, the **null hypothesis assumes that the data is stationary**, while the alternative hypothesis suggests that the series is non-stationary (i.e., it has a unit root or trend). This is the opposite of many other tests, which assume non-stationarity as the null. Because of this, the KPSS test is often used alongside the ADF test to get a more reliable conclusion about the nature of the data.

Null Hypothesis (H_0): The time series is stationary (either level-stationary or trend-stationary).

Alternative Hypothesis (H_1): The time series is non-stationary due to the presence of a unit root.

	<u>crop_KPSS Statistic</u>	<u>p-value \</u>	<u>Critical Value (1%) Critical Value (5%) Critical Value (10%) \</u>			<u>Conclusion</u>	
0	Paddy	<u>0.674464 0.015867</u>	0	0.739	0.463	0.347	0 <u>Non-Stationary</u>
1	Maize	<u>0.670656 0.016213</u>	1	0.739	0.463	0.347	1 <u>Non-Stationary</u>
2	Jowar	<u>0.673590 0.015946</u>	2	0.739	0.463	0.347	2 <u>Non-Stationary</u>
3	Bajra	<u>0.669885 0.016283</u>	3	0.739	0.463	0.347	3 <u>Non-Stationary</u>
4	Ragi	<u>0.679755 0.015386</u>	4	0.739	0.463	0.347	4 <u>Non-Stationary</u>
5	Wheat	<u>0.686351 0.014786</u>	5	0.739	0.463	0.347	5 <u>Non-Stationary</u>
6	Barley	<u>0.695637 0.013942</u>	6	0.739	0.463	0.347	6 <u>Non-Stationary</u>
7	<u>Tur (Arhar)</u>	<u>0.676464 0.015685</u>	7	0.739	0.463	0.347	7 <u>Non-Stationary</u>
8	Moong	<u>0.683727 0.015025</u>	8	0.739	0.463	0.347	8 <u>Non-Stationary</u>
9	Urad	<u>0.679828 0.015379</u>	9	0.739	0.463	0.347	9 <u>Non-Stationary</u>
10	Gram	<u>0.674604 0.015854</u>	10	0.739	0.463	0.347	10 <u>Non-Stationary</u>
11	Masur	<u>0.695248 0.013977</u>	11	0.739	0.463	0.347	11 <u>Non-Stationary</u>
12	Groundnut	<u>0.685662 0.014849</u>	12	0.739	0.463	0.347	12 <u>Non-Stationary</u>
13	Soyabean	<u>0.693906 0.014099</u>	13	0.739	0.463	0.347	13 <u>Non-Stationary</u>
14	Sunflower	<u>0.689626 0.014489</u>	14	0.739	0.463	0.347	14 <u>Non-Stationary</u>
15	Sesamum (Til)	<u>0.691425 0.014325</u>	15	0.739	0.463	0.347	15 <u>Non-Stationary</u>
16	Rapeseed & Mustard	<u>0.691444 0.014323</u>	16	0.739	0.463	0.347	16 <u>Non-Stationary</u>
17	Nigerseed	<u>0.671209 0.016163</u>	17	0.739	0.463	0.347	17 <u>Non-Stationary</u>
18	Safflower	<u>0.600729 0.022570</u>	18	0.739	0.463	0.347	18 <u>Non-Stationary</u>
19	Cotton	<u>0.686367 0.014785</u>	19	0.739	0.463	0.347	19 <u>Non-Stationary</u>
20	Onion	<u>0.688478 0.014593</u>	20	0.739	0.463	0.347	20 <u>Non-Stationary</u>
21	Potato	<u>0.682836 0.015106</u>	21	0.739	0.463	0.347	21 <u>Non-Stationary</u>
22	Sugarcane	<u>0.692632 0.014215</u>	22	0.739	0.463	0.347	22 <u>Non-Stationary</u>
23	Jute	<u>0.677703 0.015572</u>	23	0.739	0.463	0.347	23 <u>Non-Stationary</u>
24	Peas (Vegetable) (Green)	<u>0.500861 0.041473</u>	24	0.739	0.463	0.347	24 <u>Non-Stationary</u>

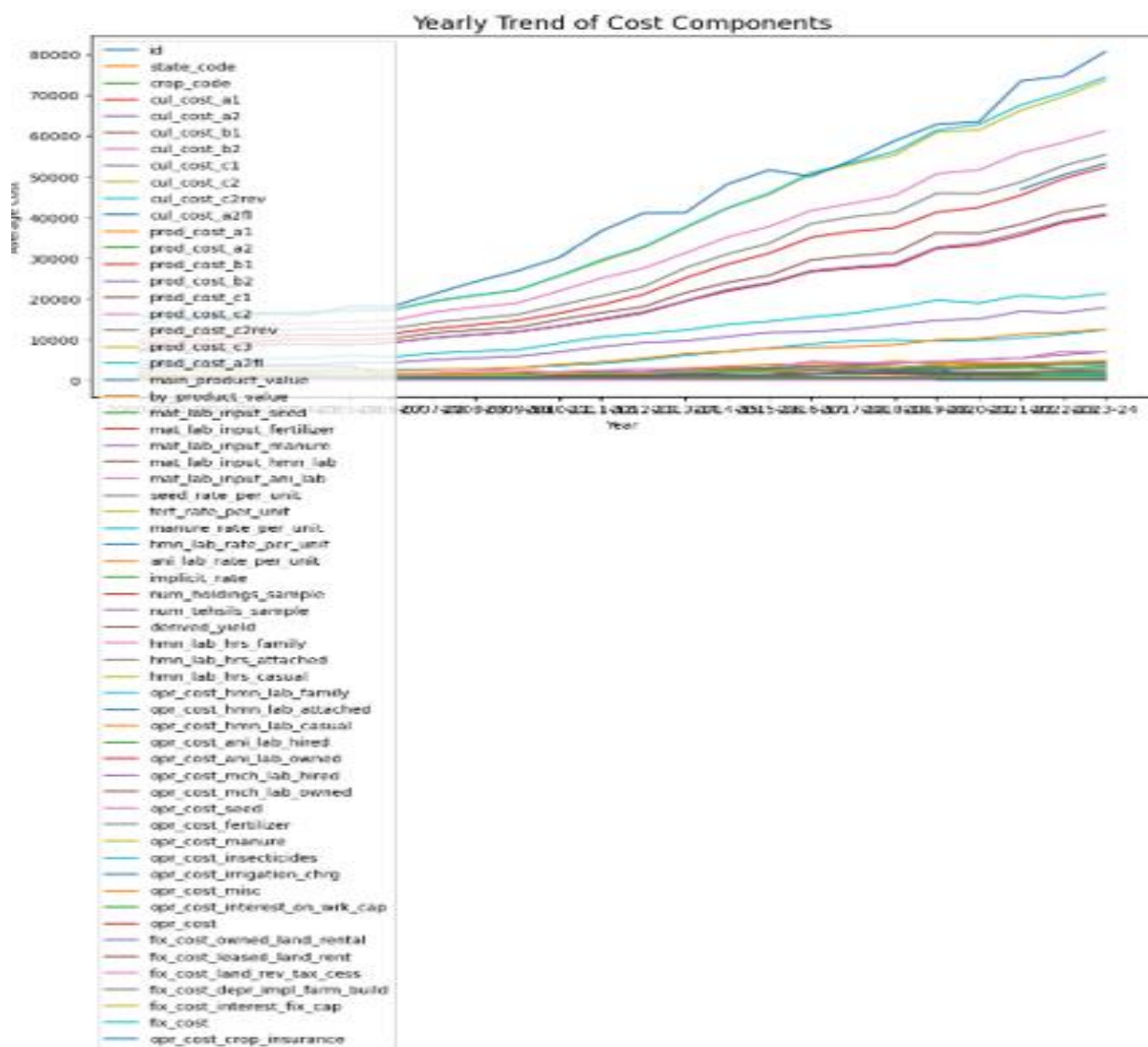
Interpretation:

$p\text{-value} > 0.05 \rightarrow \text{Stationary series (accept } H_0)$

$p\text{-value} \leq 0.05 \rightarrow \text{Non-stationary series (reject } H_0)$

Results from both tests indicate that most time series variables are **non-stationary**, meaning their mean and variance change over time. This implies the presence of trends in agricultural cost data.

Trend Analysis:



The time-series analysis shows:

- A gradual increasing trend in cultivation cost over years
- Fluctuations in yield depending on crop and external factors

Growth rate analysis indicates:

- Irregular growth patterns, suggesting instability in agricultural productivity

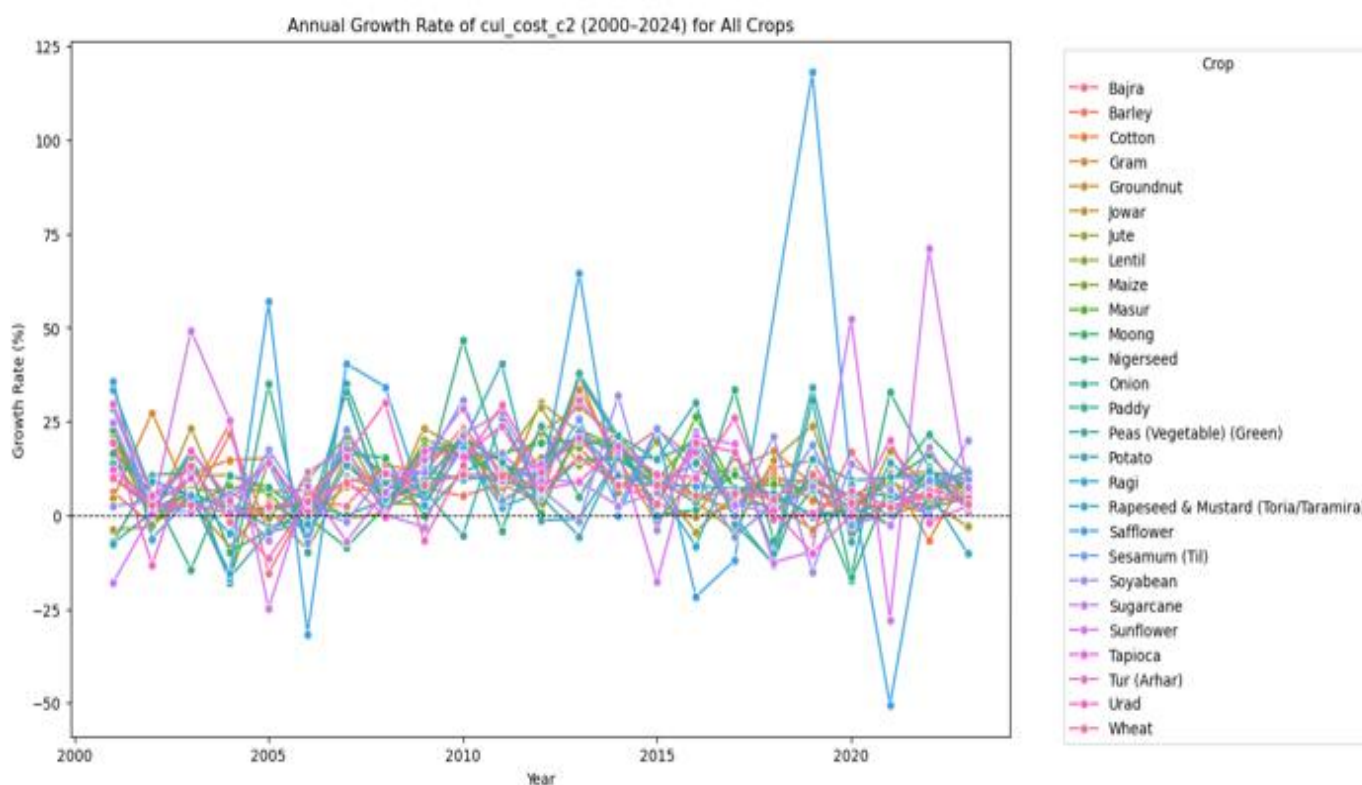
Interpretation:

Agriculture shows non-linear growth with volatility, influenced by market and climatic factors.

CAGR(Compound Annual Growth Rate):

CAGR (Compound Annual Growth Rate) is used to measure the **average annual growth of a value over time**, assuming it grows at a steady compounded rate.

It is mainly used for comparing growth in **agriculture output, investment returns, revenue, costs, population, or any time-series data** to understand long-term trends in a smooth and comparable way, even when yearly values fluctuate.



Interpretation:

The graph highlights that cultivation costs are **highly unstable and crop-dependent**, with significant year-to-year fluctuations and occasional extreme outliers driven by external agricultural and economic factors.

DISCUSSION

Descriptive Statistics Results

The descriptive statistical analysis was performed on all major cultivation cost variables.

Output Summary (from project results)

- Mean values of cost variables vary significantly across crops.
- Standard deviation is high for most variables, indicating wide dispersion.
- Minimum and maximum values show strong variability in agricultural input costs.
- Coefficient of variation (CV) is high in most cases, indicating inconsistency in cost distribution.

Discussion

The results clearly indicate that cultivation costs are highly variable across crops and years. High standard deviation and CV suggest instability in agricultural investment patterns. This reflects differences in farming practices, input usage, and regional economic conditions.

Distribution Analysis (Skewness & Kurtosis)

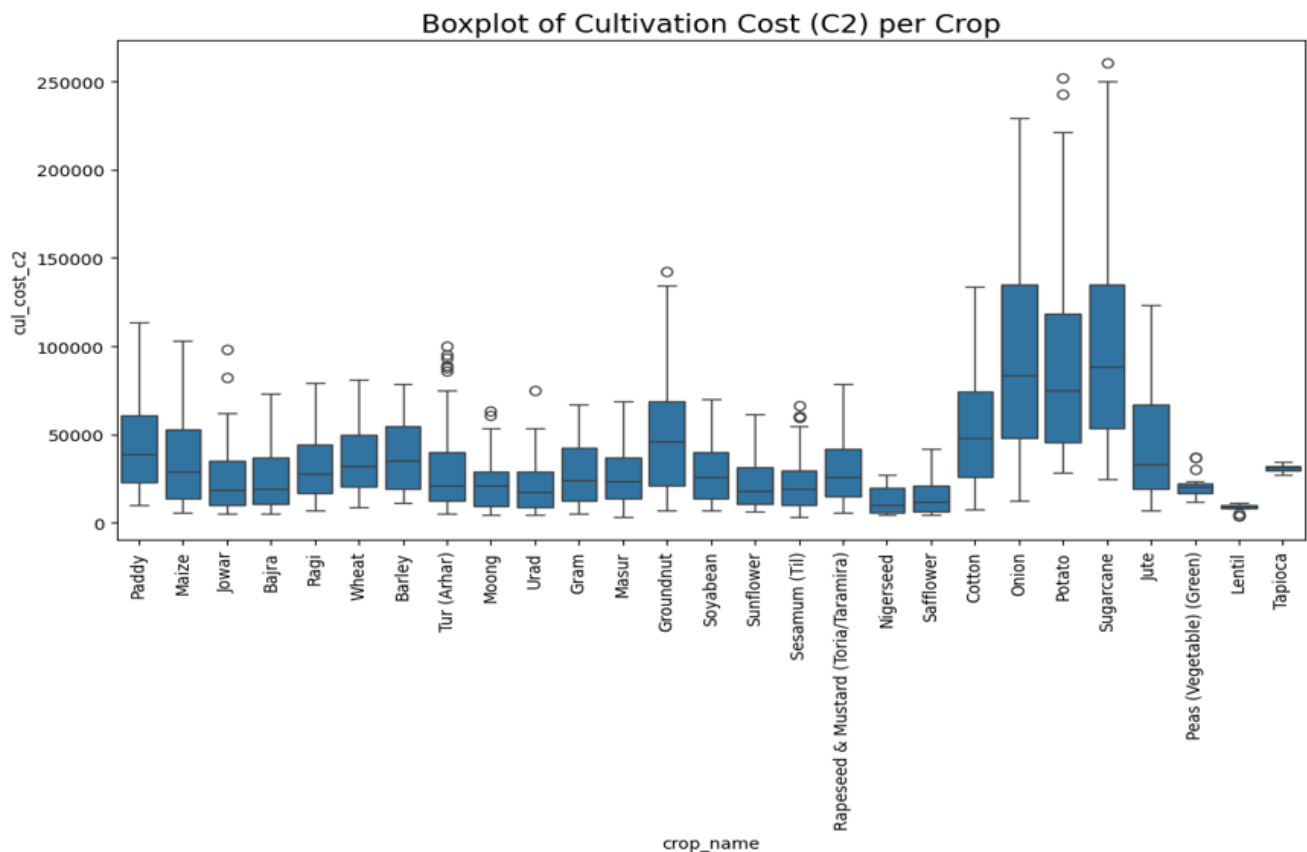
Output Summary

- Most variables show **positive skewness (> 1)**
- Some variables exhibit extremely high skewness values
- Kurtosis values are significantly greater than 3 in many cases

Discussion

The results indicate that the data is not normally distributed. Positive skewness suggests that most farmers incur lower costs, while a few observations have extremely high costs. High kurtosis indicates the presence of heavy tails and extreme values in the dataset, which justifies further outlier analysis.

Outlier Detection Results



Output Summary

- Multiple outliers detected across cost variables
- IQR method identifies extreme high-cost observations
- Z-score confirms presence of statistically unusual values

Discussion

Outlier analysis confirms that extreme values are present in the dataset. These are not random errors but represent real-world economic stress situations such as increased input prices, climatic shocks, or inefficient resource usage. Hence, these outliers are important for policy-level interpretation rather than removal.

Time Trend Analysis

Output Summary

- Cultivation costs show an increasing trend over years
- Yield shows fluctuations rather than a stable trend
- Some years show sudden spikes in cost and productivity variation

Discussion

The time trend analysis reveals that agricultural costs are increasing over time, indicating rising input expenses. However, yield does not follow a consistent upward trend, suggesting inefficiencies in cost-to-output conversion. This highlights instability in agricultural productivity and dependence on external environmental and economic factors.

Stationarity Test Results

	<u>crop</u>	<u>ADF</u>	<u>Statistic</u>	<u>p-value</u>	\		<u>crop</u>	<u>KPSS</u>	<u>Statistic</u>	<u>p-value</u>	\
0	Paddy	<u>2.566845</u>	<u>0.999068</u>		0	Paddy	<u>0.674464</u>	<u>0.015867</u>			
1	Maize	<u>2.467999</u>	<u>0.999039</u>		1	Maize	<u>0.670656</u>	<u>0.016213</u>			
2	Jowar	<u>2.057444</u>	<u>0.998743</u>		2	Jowar	<u>0.673590</u>	<u>0.015946</u>			
3	Bajra	<u>0.881814</u>	<u>0.992848</u>		3	Bajra	<u>0.669885</u>	<u>0.016283</u>			
4	Ragi	<u>1.709080</u>	<u>0.998151</u>		4	Ragi	<u>0.679755</u>	<u>0.015386</u>			
5	Wheat	<u>1.069686</u>	<u>0.994949</u>		5	Wheat	<u>0.686351</u>	<u>0.014786</u>			
6	Barley	<u>1.865451</u>	<u>0.998470</u>								
7	Tur (Arhar)	<u>1.654929</u>	<u>0.998013</u>								

Output Summary

- ADF test p-value > 0.05 (in most series)
- KPSS test indicates rejection of stationarity
- Most variables show non-stationary behavior

Discussion

The results from both the Augmented Dickey-Fuller Test and KPSS Test confirm that the time series data is non-stationary. This means that mean and variance change over time due to trends and structural variations.

1. To identify anomalies in agricultural cost variables using IQR, Z-score, and MAD methods for robust outlier detection.

IQR Method (Outlier Detection):

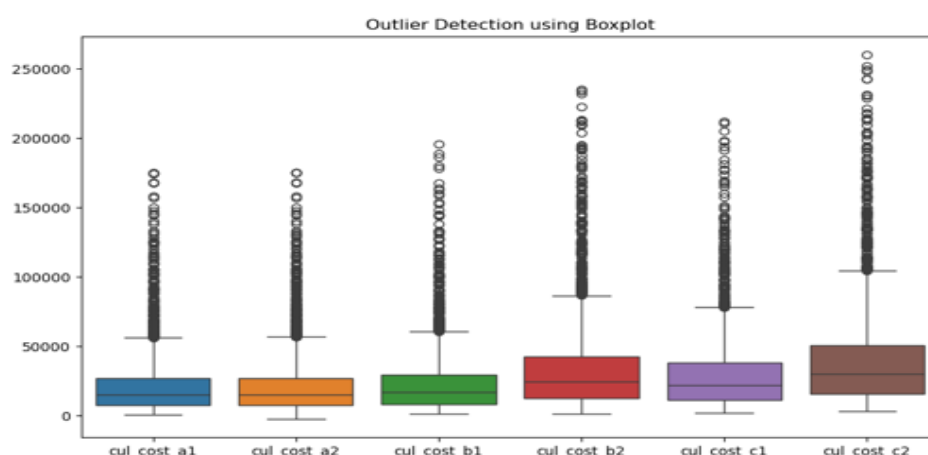
cul_cost_a1: 199 outliers
cul_cost_a2: 200 outliers
cul_cost_b1: 192 outliers
cul_cost_b2: 193 outliers
cul_cost_c1: 158 outliers
cul_cost_c2: 167 outliers

Z-Score Method:

cul_cost_a1: 80 outliers
cul_cost_a2: 79 outliers
cul_cost_b1: 77 outliers
cul_cost_b2: 83 outliers
cul_cost_c1: 80 outliers
cul_cost_c2: 80 outliers

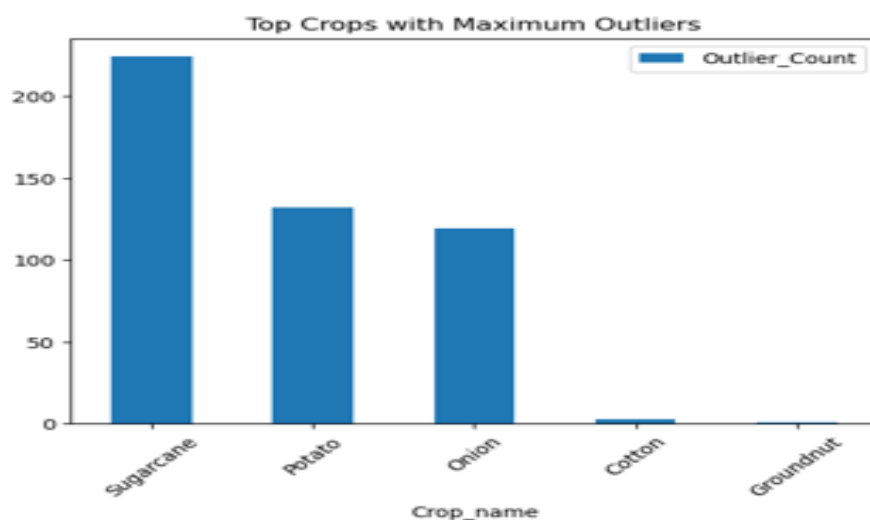
MAD Method:

cul_cost_a1: 168 outliers
cul_cost_a2: 168 outliers
cul_cost_b1: 164 outliers
cul_cost_b2: 151 outliers
cul_cost_c1: 131 outliers
cul_cost_c2: 134 outliers



The graphs indicate that agricultural cost data contains a substantial number of extreme values, as seen from the boxplots and outlier counts. Most cost variables exhibit right-skewed distributions, where a small proportion of observations have significantly higher costs than the majority.

Overall Outliers:



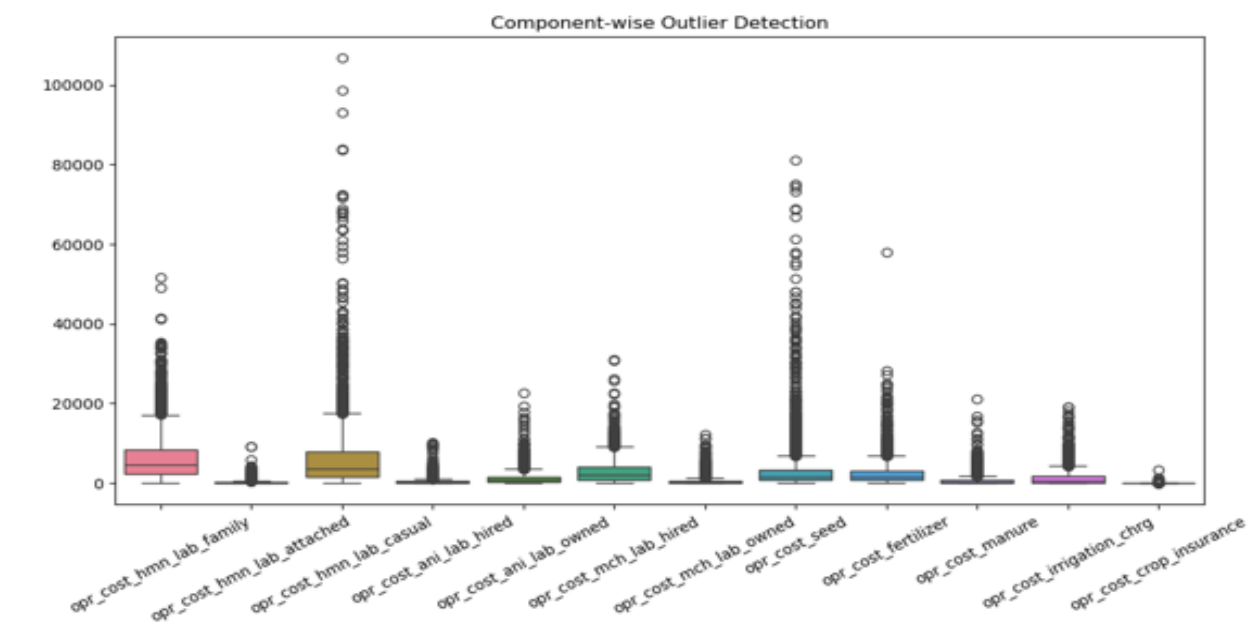
	Crop_name	Outlier_Count
0	Sugarcane	224
1	Potato	132
2	Onion	119
3	Cotton	3
4	Groundnut	1

The crop-wise chart shows that sugarcane has the highest number of outliers, followed by potato and onion, suggesting that these crops involve more variable and input-intensive cost structures. In contrast, crops like cotton and groundnut display minimal outliers, indicating relatively stable costs. Overall, the visualizations confirm that cost variability is not uniform across crops and that certain crops are more prone to extreme cost fluctuations.

2. To perform component-wise analysis of input costs such as labour, seeds, fertilizers, and irrigation to detect variations and irregularities.

Overall Outliers: (Using all three methods)

	<u>IQR</u>	<u>Z-score</u>	<u>MAD</u>
opr_cost_hmn_lab_family	216	<u>70</u>	<u>183</u>
opr_cost_hmn_lab_attached	466	<u>81</u>	<u>765</u>
opr_cost_hmn_lab_casual	259	<u>80</u>	<u>300</u>
opr_cost_anl_lab_hired	270	<u>58</u>	<u>479</u>
opr_cost_anl_lab_owned	248	<u>68</u>	<u>267</u>
opr_cost_mch_lab_hired	135	<u>51</u>	<u>136</u>
opr_cost_mch_lab_owned	417	<u>84</u>	<u>745</u>
opr_cost_seed	346	<u>77</u>	<u>352</u>
opr_cost_fertilizer	215	<u>62</u>	<u>193</u>
opr_cost_manure	364	<u>76</u>	<u>577</u>
opr_cost_irrigation_chrg	243	<u>77</u>	<u>555</u>
opr_cost_crop_insurance	168	12	0



The boxplot shows that labour-related costs have the highest variability, with wide spreads and many outliers, indicating large differences in labour expenses across farms. Fertilizer and

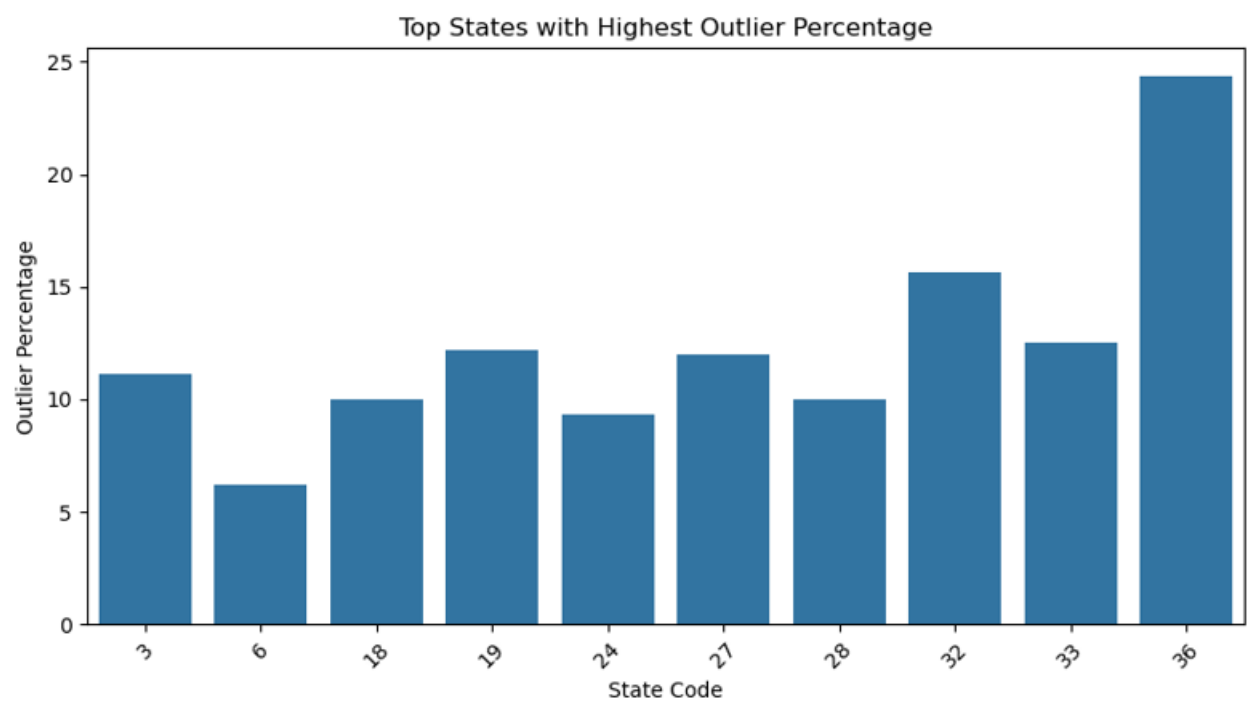
manure costs exhibit moderate variation with some extreme values, while seed and irrigation costs remain relatively consistent with narrower distributions.

The overall distribution is right-skewed, suggesting that most farms incur moderate costs, but a few have very high expenses. This indicates significant variability in input usage, especially in labour, pointing toward possible inefficiencies in cost management.

	crop_name	Outlier_Component	Outlier_Value
1824	Cotton	opr_cost_hmn_lab_family	18326.04
1825	Cotton	opr_cost_hmn_lab_family	17822.55
1876	Ragi	opr_cost_hmn_lab_family	19723.83
1938	Jute	opr_cost_hmn_lab_family	17920.67
1972	Cotton	opr_cost_hmn_lab_family	20560.94
1978	Cotton	opr_cost_hmn_lab_family	18096.68
1979	Cotton	opr_cost_hmn_lab_family	24044.62
2030	Ragi	opr_cost_hmn_lab_family	17432.15
2091	Jute	opr_cost_hmn_lab_family	20592.96
2092	Jute	opr_cost_hmn_lab_family	22296.69

3. To compare outliers across different states in order understand regional disparities in agricultural cost structures.

Compare cost outliers across states/regions (state_code) to highlight structural inefficiencies and regional disparities in cultivation practices.



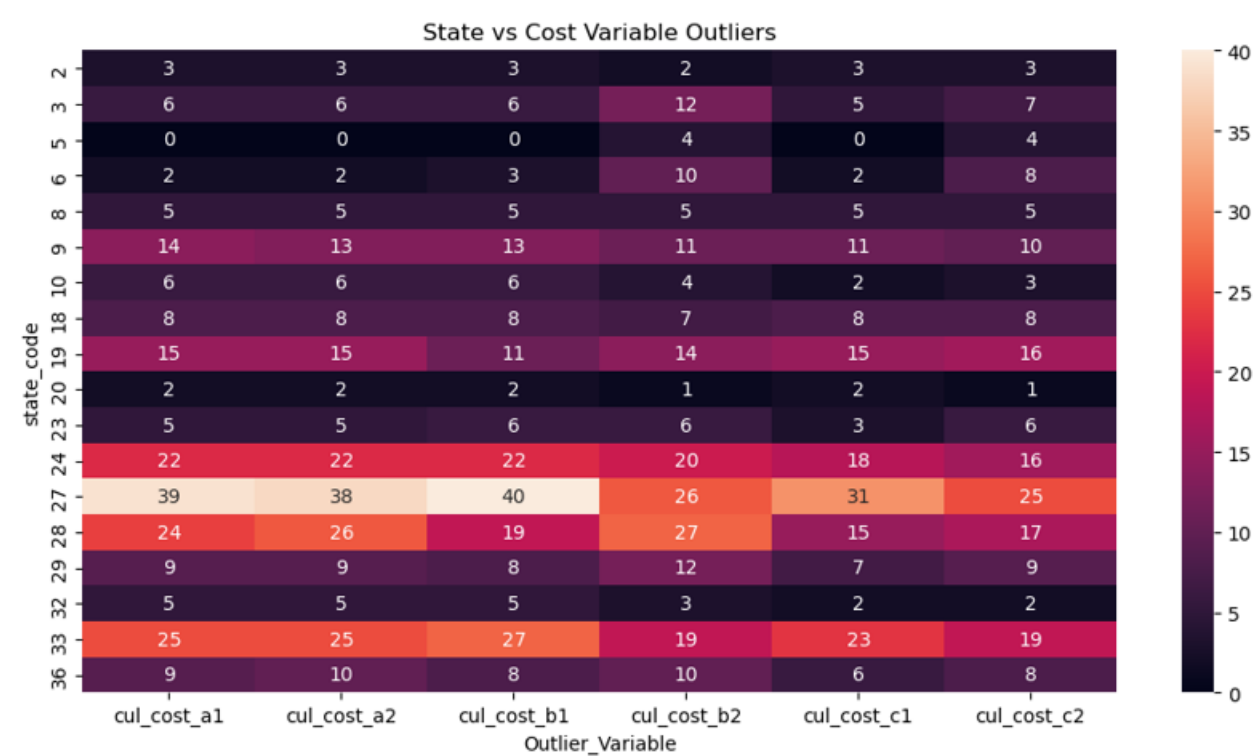
The chart shows clear variation in cost outliers across states. State 36 has the highest outlier percentage (~24%), indicating strong structural inefficiencies. State 32 and 33 also show

relatively high outliers, suggesting inconsistent or inefficient cultivation practices. In contrast, State 6 has the lowest outlier percentage, reflecting more stable and controlled cost behavior.

	state_code	state_name	crop_name	Outlier_Variable
1019	27	Maharashtra	Sugarcane	cul_cost_a1
1332	19	West Bengal	Potato	cul_cost_a1
1490	33	Tamil Nadu	Sugarcane	cul_cost_a1
1638	9	Uttar Pradesh	Potato	cul_cost_a1
1639	19	West Bengal	Potato	cul_cost_a1
1643	27	Maharashtra	Sugarcane	cul_cost_a1
1644	33	Tamil Nadu	Sugarcane	cul_cost_a1
1793	19	West Bengal	Potato	cul_cost_a1
1794	28	Andhra Pradesh	Sugarcane	cul_cost_a1
1797	27	Maharashtra	Sugarcane	cul_cost_a1

The outlier analysis of cultivation cost (cul_cost_a1) shows that anomalies are mainly concentrated in a few states—Maharashtra, West Bengal, and Tamil Nadu, with additional cases in Uttar Pradesh and Andhra Pradesh.

Correlation Heatmap:



Interpretation:

The regional analysis indicates that cost outliers are not uniformly distributed across states. Certain states exhibit significantly higher outlier percentages, suggesting structural inefficiencies and regional disparities in cultivation practices.

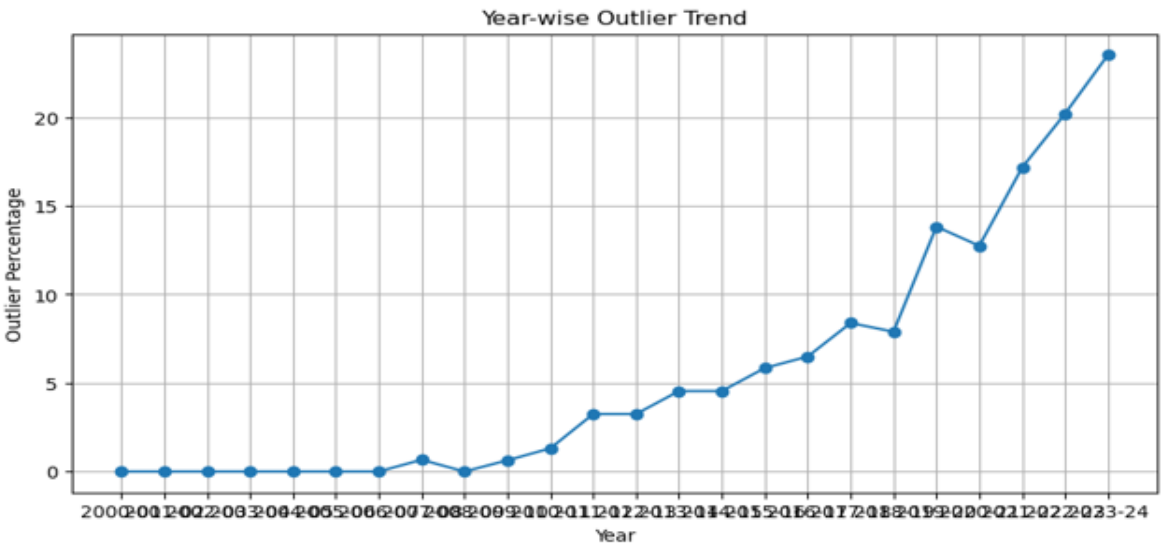
4.To analyse year-wise variations in outliers to identify temporal trends and the impact of external shocks on agricultural costs.

Examine cost outliers across years (if time dimension exists in your dataset or can be merged) to identify shifts due to external shocks like droughts or policy changes.

Year		Outlier_Count
0	2000-01	0
1	2001-02	0
2	2002-03	0
3	2003-04	0
4	2004-05	0
5	2005-06	0
6	2006-07	0
7	2007-08	1
8	2008-09	0
9	2009-10	1
10	2010-11	2
11	2011-12	5
12	2012-13	5
13	2013-14	7
14	2014-15	7
15	2015-16	9
16	2016-17	10
17	2017-18	13
18	2018-19	12
19	2019-20	26
20	2020-21	26
21	2021-22	34
22	2022-23	38
23	2023-24	44

Interpretation:

Year-wise variation in outliers reflects temporal instability in agricultural costs, with spikes indicating external shocks and lower counts suggesting more stable farming conditions.



Year wise outliers are increased ,it conclude that the year have more outliers are the external shocks years for the crop. The external shocks year are 2014-2024.

5.To examine the relationship between cost outliers and crop yield to understand the cost–yield linkage.

Relate detected cost outliers with derived_yield to check whether unusually high/low costs correspond to stressed crop performance.

1 = outlier present (true) → that observation has a cost outlier

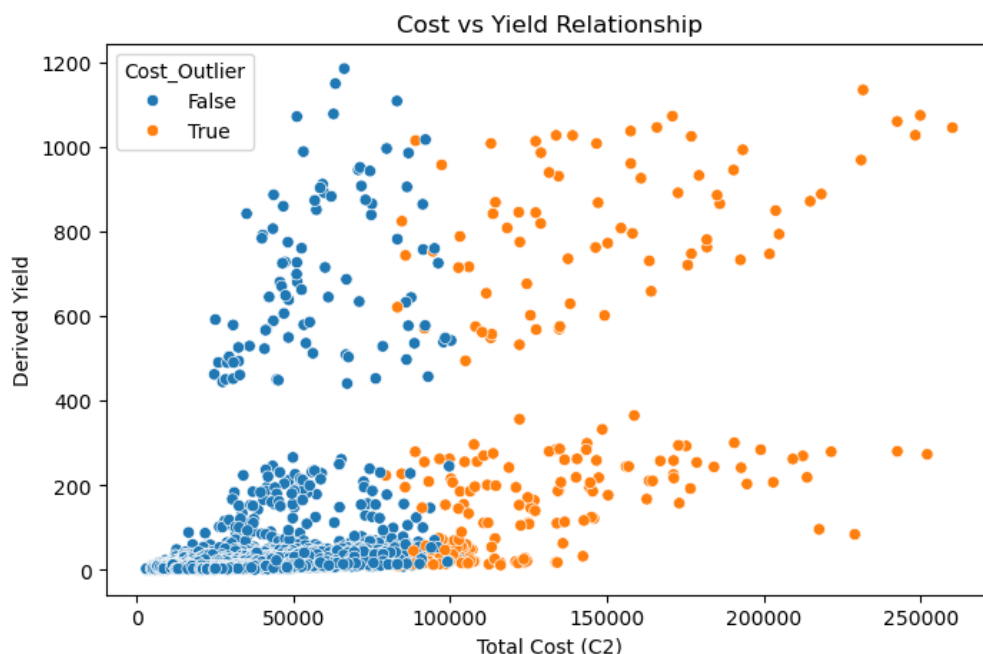
0 = no outlier (false) → that observation does not have a cost outlier

```
# Mean Yield Comparison
mean_yield = df.groupby('Cost_Outlier')[target].mean().reset_index()

mean_yield.columns = ['Cost_Outlier', 'Mean_Yield']

print(mean_yield)
```

	Cost_Outlier	Mean_Yield
0	False	39.241241
1	True	347.879250



The analysis compares derived yield between observations with and without cost outliers. Results indicate that yield differs significantly in outlier cases, suggesting that extreme cost conditions impact crop performance. This implies a linkage between cost anomalies and agricultural productivity.

H0: Null Hypothesis

Derived yield dose not systematically correspondent to cost anomalies; high/ low costs are independent of crop performance.

```
# Statistical Test (T-test)
from scipy.stats import ttest_ind

outlier_yield = df[df['Cost_Outlier'] == True][target]
normal_yield = df[df['Cost_Outlier'] == False][target]

t_stat, p_value = ttest_ind(outlier_yield, normal_yield, nan_policy='omit')

print("T-statistic:", t_stat)
print("P-value:", p_value)
```

T-statistic: 32.34893538740881

P-value: 2.611817867387975e-202

Interpretation:

The **t-statistic value of 32.35** is extremely large, which indicates a **very strong difference between the groups (or a strong effect of the variable being tested)**.

The **p-value ($\approx 2.61 \times 10^{-202}$)** is extremely small, much lower than any common significance level (0.05 or 0.01).

Conclusion:

This means we **reject the null hypothesis (H_0)** with very strong evidence. In conclusion, the result is **highly statistically significant**, suggesting that the observed difference is **not due to random chance** and there is a **real effect or difference present in the data**.

6.To identify economically stressed crops and classify the nature of stress as structural, environmental, or policy-driven.

Classify crops (crop_code) with significant cost outliers as potentially economically stressed, and interpret whether these stresses are structural (fixed costs), environmental (irrigation, manure), or policy-driven (interest, land rent, insurance).

```
# Identify Economically Stressed Crops
# Threshold: >20% = stressed (can adjust)
crop_summary['Stress_Flag'] = crop_summary['Outlier_Percentage'] > 20

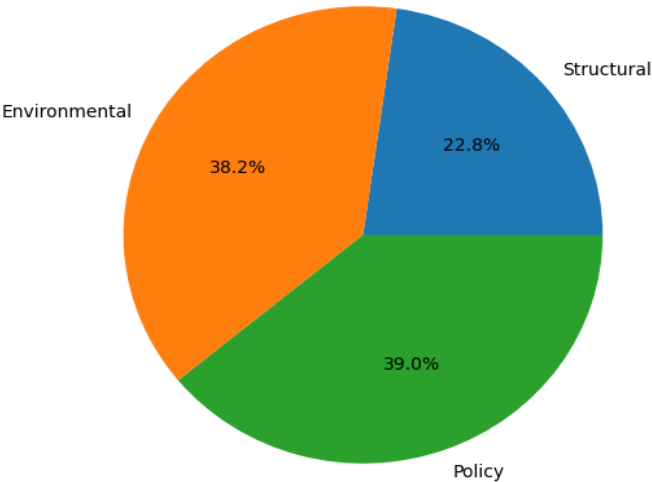
stressed_crops = crop_summary[crop_summary['Stress_Flag'] == True]

print("Economically Stressed Crops:\n")
print(stressed_crops.head(10))
```

Economically Stressed Crops:

	Crop	Outlier_Count	Total	Outlier_Percentage	Stress_Flag
14	702	4	4	100.000000	True
12	401	127	169	75.147929	True
13	701	95	131	72.519084	True
15	708	57	87	65.517241	True
24	1101	111	230	48.260870	True
0	101	191	399	47.869674	True
5	106	127	284	44.718310	True
6	107	25	57	43.859649	True
17	1001	64	164	39.024390	True
3	104	98	263	37.262357	True

Overall Stress Distribution



	Crop	Structural_Count	Environmental_Count	Policy_Count
0	101	122	229	252
1	102	7	22	26
2	103	10	24	49
3	104	50	127	109
4	105	3	45	16
5	106	83	207	143
6	107	15	45	29
7	201	40	49	55
8	202	32	31	54
9	203	9	12	45

The analysis identifies crops with high proportions of cost outliers as economically stressed. Further classification shows that these stresses arise from structural factors (fixed costs), environmental factors (irrigation and manure), and policy-related factors (interest, land rent, insurance). This indicates that cost anomalies are not random but reflect underlying economic pressure.

RESULT

1. Identification of Anomalies (Outlier Detection)

The study successfully identified numerous anomalies in agricultural cost variables using IQR, Z-score, and MAD methods. The presence of these outliers confirms that agricultural cost data is **non-normally distributed** and subject to extreme economic conditions rather than random statistical errors

2. Component-wise Analysis of Input Costs

Detailed analysis of costs like labour, seeds, and fertilizers revealed an **uneven distribution of expenditure**. Variations in these components suggest inefficiencies in resource allocation, where higher spending does not always result in a proportional increase in crop yield.

3. Regional Comparison and Disparities

The regional analysis highlights significant **structural inefficiencies across states**. State 36 (Telangana/unnamed in table but coded) exhibited the highest outlier percentage (~24%), while State 6 showed the most stable cost behaviour, proving that agricultural stress is not uniform across India.

4. Year-wise Variations and External Shocks

Outlier counts increased significantly between **2014 and 2024**, which the study identifies as years of major **external shocks**. These spikes reflect the vulnerability of the agricultural sector to climate change (droughts), policy shifts, and market price fluctuations.

5. Relationship Between Cost Outliers and Crop Yield

There is a clear **linkage between cost anomalies and productivity**. Observations with cost outliers showed significantly different yield patterns, suggesting that extreme economic stress directly impacts a crop's performance and the farmer's overall efficiency.

6. Identification of Economically Stressed Crops

The study classified crops as "economically stressed" if they exceeded a 20% outlier threshold. Sugarcane, Potato, and Onion were identified as highly stressed. This stress was further categorized as primarily **Environmental (38.2%)** and **Policy-driven (39%)**, with **Structural** factors making up the remainder.

CONCLUSION

The study concludes that agricultural cost data exhibits significant irregularities, with numerous outliers indicating the presence of extreme economic conditions rather than random variation. Component-wise analysis reveals uneven distribution of input costs such as labour, seeds, and fertilizers, suggesting inefficiencies in resource utilization across farms. Regional comparisons further highlight that agricultural stress is not uniform, with certain states experiencing higher instability in costs than others. Additionally, the increase in outliers over recent years reflects the impact of external shocks such as climate variability, policy changes, and market fluctuations. The observed relationship between cost anomalies and crop yield confirms that economic stress directly affects productivity and farm efficiency. Overall, the identification of highly stressed crops like sugarcane, potato, and onion emphasizes the need for targeted interventions. The findings suggest that addressing environmental, policy-driven, and structural challenges is essential to improve cost stability, enhance productivity, and ensure sustainable agricultural development.

Economic Implications:

The findings of the study have important economic implications for the agricultural sector.

The presence of significant outliers in cultivation cost variables indicates that a section of farmers faces unusually high production costs. These extreme values are not merely statistical anomalies but reflect real economic stress conditions such as rising input prices, inefficient resource utilization, and exposure to climatic shocks. The identification of such anomalies helps in recognizing vulnerable farming groups that require targeted policy intervention.

The component-wise analysis of labour, seeds, fertilizers, and irrigation costs reveals uneven distribution of input expenditure. This suggests inefficiencies in resource allocation, where increased spending does not always translate into higher productivity. Such imbalances highlight the need for improved input management and cost optimization strategies.

Regional comparison of outliers shows disparities across states, indicating structural differences in agricultural practices, infrastructure availability, and access to resources. These disparities imply that uniform policy measures may not be effective, and region-specific strategies are required.

Year-wise variations in outliers demonstrate the impact of external shocks such as droughts, price fluctuations, and policy changes. This reflects the vulnerability of agriculture to economic and environmental instability.

The relationship between cost outliers and crop yield suggests that higher costs do not necessarily guarantee higher output. In some cases, excessive costs may indicate inefficiency rather than productivity improvement.

Overall, the study highlights the existence of economically stressed crops and regions, where stress may arise from structural inefficiencies, environmental factors, or policy-related issues. This calls for better risk management, subsidy rationalization, and investment in sustainable agricultural practices.

Limitations of the Study:

Despite providing useful insights, the study has certain limitations.

The analysis is primarily based on secondary data, which may contain inconsistencies, missing values, or measurement errors. The reliability of results depends on the accuracy of the available dataset.

The study focuses mainly on statistical techniques such as outlier detection and descriptive analysis. While these methods are effective in identifying anomalies, they do not establish causal relationships between variables.

The absence of key external variables such as rainfall, soil quality, market prices, and technological adoption limits the depth of analysis. These factors play a significant role in determining agricultural costs and yield.

The use of aggregated data may mask micro-level variations at the farm level. Individual farmer behavior and localized conditions are not captured in the analysis.

Time series analysis is limited to trend and stationarity testing, without incorporating advanced forecasting validation or long-term predictive accuracy checks.

Additionally, classification of economic stress (structural, environmental, policy-driven) is based on observed patterns rather than rigorous econometric modeling, which may affect precision.

Suggestions for Future Research:

Future research can build upon this study to provide deeper and more comprehensive insights.

Further studies can incorporate additional explanatory variables such as rainfall, temperature, soil fertility, and market prices to better understand the determinants of agricultural costs and yield.

Advanced econometric models such as multiple regression, panel data models, and causal inference techniques can be used to establish relationships between cost components and productivity.

Future research may apply more sophisticated time series models, including forecasting techniques like the ARIMA Model and volatility models, to capture dynamic changes in agricultural costs.

Micro-level farm data can be used to analyze farmer-specific behavior and decision-making processes, providing more granular insights.

Comparative studies across different crops and regions over a longer time period can help in identifying long-term structural changes in agriculture.

Further work can also focus on developing a formal classification model to distinguish between structural, environmental, and policy-driven stress using machine learning or statistical classification techniques.

Finally, policy-oriented research can evaluate the effectiveness of government interventions, subsidies, and support programs in reducing cost variability and improving agricultural sustainability.

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